

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. (EE) Examination

April 2013

EE 408 Power System Stability & Control

Date: 24.04.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Derive the swing equation of synchronous machine [07]
- Q - 2 (a) Explain the equal area criteria for the power system stability analysis with the case of sudden application of mechanical input [07]
- (b) For the system shown in Figure 1, a balanced three phase fault occurs at the mid point of line 2 when the power transfer is 1.5 pu in the system. $E = 1.2$, $V = 1$, $X_d' = 0.2$, $X_1 = X_2 = 0.4$ pu. Determine whether the system is stable for a sustained fault? Also Find the critical clearing angle for this system [07]

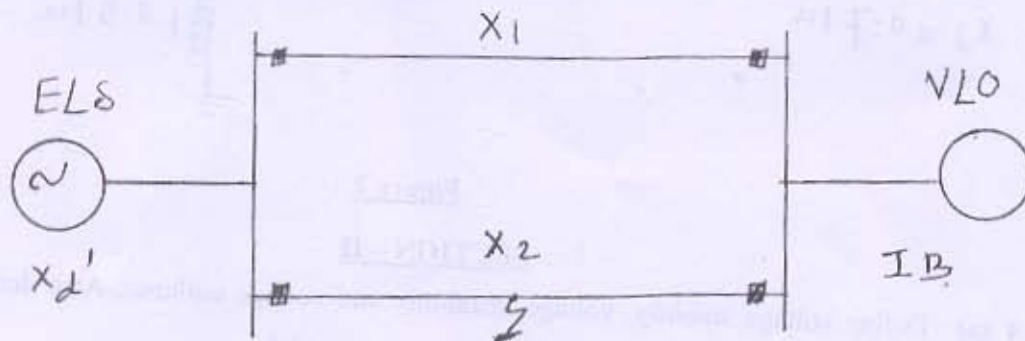


Figure 1

OR

- Q - 2 (a) Derive the synchronizing torque coefficient and state how the stability can be analyzed with it. [07]
- (b) A 200 MVA, 11 kV, 50 Hz, 4 pole turbo generator has an inertia constant of 6 MJ/MVA. (a) Find the stored energy in rotor at synchronous speed (b) Machine is operating at a load of 120 MW when load suddenly increases to 160 MW. Find the rotor retardation. Neglect losses. (c) If the retardation is maintained for 5 cycles, find the change in power angle and rotor speed in rpm at the end of this period. [07]

Q - 3 (a) Explain point by point method for the solution of the swing equation? [07]

(b) Explain the methods of power system stability improvement in details [07]

OR

Q - 3 (a) Explain the transient stability analysis for the multi machine system [07]

(b) For the system shown in Figure 2, find [07]

(a) transfer reactance and steady state power transfer limit when inductor switch is open

(b) transfer reactance and steady state power transfer limit when inductor switch is closed

(c) transfer reactance and steady state power transfer limit when inductor is replaced by capacitor of same value

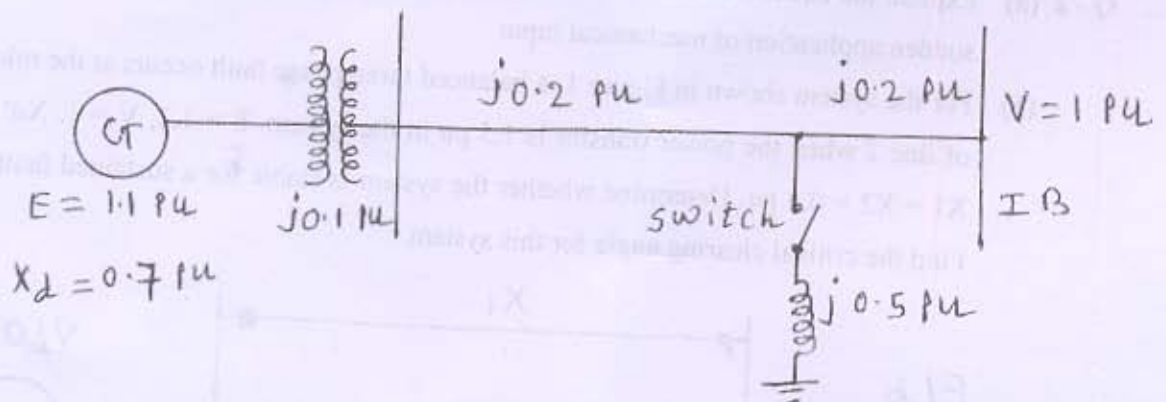


Figure 2

SECTION - II

Q - 4 (a) Define voltage stability, voltage instability and voltage collapse. Also derive active and reactive power transmission by elementary model [07]

Q - 5 (a) Derive the expression for voltage stability limit for the power system [07]

(b) A short transmission line is being fed by a source having a source reactance of 0.22 pu on 1000 MW base. The line reactance is 0.15 pu on 1000 MW and rated voltage base. The line is supplying 1 puMW at 0.8 lagging PF at rated voltage. What will be the voltage at the receiving end if the receiving end line breaker opens suddenly. [07]

OR

Q - 5 (a) Explain block diagram representation of SMIB system using classical model [07]

(b) Write a short on power system stabilizer [07]

- Q - 6 (a) Explain the V – I characteristic of SVC and STATCOM and make the comparison [07]
between two
- (b) Explain midpoint voltage regulation for increasing the steady state power transfer [07]
capability and transient stability margin using SVC

OR

- Q - 6 (a) State the advantages of TCSC. Explain the principle of TCSC operation [05]
- (b) Explain Any Three [09]
- [1] Discuss the corrective measures to avoid the voltage instability
 - [2] List out the sources and sinks of the reactive power and state the equations of reactive power produced or consumed by them
 - [3] How series compensation can increase the transient and voltage stability?
 - [4] Why do we require FACTS devices? Give the classification of FACTS devices